

Oracle provides robust backup and recovery mechanisms to ensure data protection and availability. A well-planned backup and recovery strategy is critical for minimizing data loss and downtime in the event of failures. Below are the key components and strategies for Oracle backup and recovery:

1. Types of Backups

Oracle supports two main types of backups:

- **Physical Backups:** Physical backups are copies of physical database files. For example, a physical backup might copy database content from a local disk drive to another secure location. These are backups of the physical database files (data files, control files, and redo logs).
- **Logical Backups:** A logical backup copies data, but not physical files, from one location to another. A logical backup is used to move or archive a database, tables, or schemas and to verify database structures. These are backups of logical database objects (tables, schemas, or entire databases) using tools like **Data Pump**.

2. Backup Strategies

a. RMAN (Recovery Manager)

RMAN is Oracle's recommended tool for physical backups. It provides features like incremental backups, compression, and encryption. In Oracle, "RMAN" stands for "Recovery Manager," which is a utility that allows users to efficiently back up, restore, and recover Oracle databases, essentially providing a comprehensive data protection mechanism for the database system; it is designed to work closely with the Oracle server to detect block-level corruption during backup and restore operations.

Key points about RMAN:

- **Function:**

Primarily used for backing up and recovering database files, including tablespaces, datafiles, control files, and archived redo logs.

- **Integration with Oracle:**

RMAN is a built-in feature of the Oracle database server, meaning it doesn't require a separate installation.

- **Metadata storage:**

RMAN stores metadata about its operations in the database control file, and optionally in a dedicated recovery catalog schema for easier management.

- **Command-line access:**

Users can interact with RMAN through command-line commands, allowing for scripting and automation of backup and recovery procedures.

- **Full Backup:**

```
RMAN> BACKUP DATABASE;
```

Backs up all data files, control files, and server parameter file (SPFILE).

- **Incremental Backup:**

```
RMAN> BACKUP INCREMENTAL LEVEL 1 DATABASE;
```

Backs up only the blocks that have changed since the last backup.

- **Archived Redo Log Backup:**

RMAN> BACKUP ARCHIVELOG ALL;

Backs up all archived redo logs.

- **Control File Backup:**

RMAN> BACKUP CURRENT CONTROLFILE;

Backs up the control file.

- **Backup Validation:**

RMAN> VALIDATE BACKUPSET;

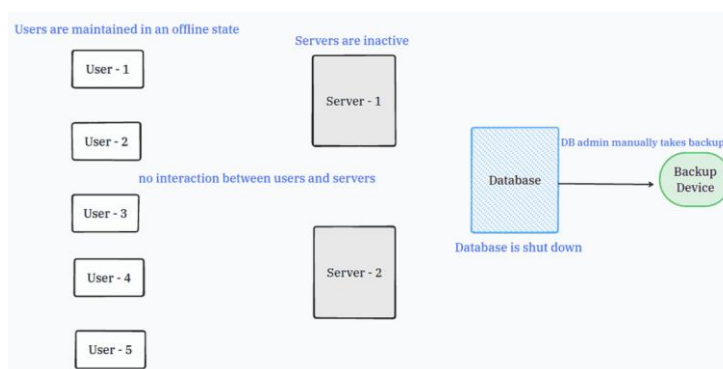
Validates the integrity of backups.

b. User-Managed Backups

In Oracle, a "User-Managed Backup" refers to a method of backing up and restoring a database by manually executing operating system commands and SQL statements, essentially managing the backup process without using Oracle's built-in Recovery Manager (RMAN) tool, which is considered the preferred method for automated database backups; meaning the user is fully responsible for creating and maintaining their database backups using traditional file system operations.

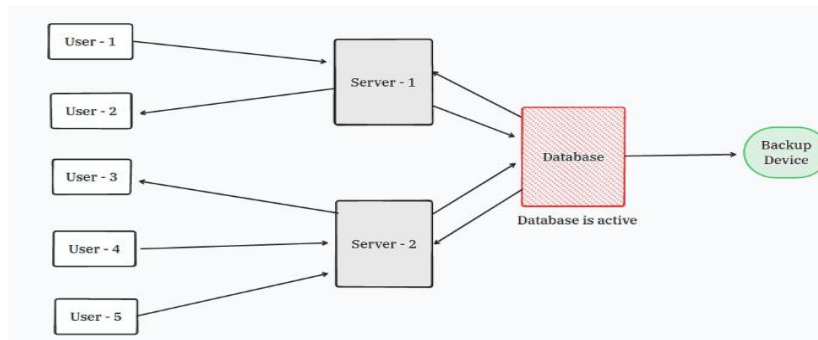
The basic user-managed procedure for recovering from a media failure is as follows:

1. Restore database file backups with operating system commands.
 2. Recover restored datafiles with the SQL*Plus RECOVER statement.
 3. If the database is closed, then open it for normal use; if it is open, then bring the recovered tablespaces back online.
- **Cold Backup:** A cold data backup in Oracle refers to the process of backing up the database when it is offline or shut down. It's also known as offline backup. During a cold backup, all database files, including data files, control files, and archived redo logs are copied while the database is not running. So it ensures that the backup represents a consistent snapshot of the database at the time the backup was initiated.
 - Shut down the database.
 - Copy all data files, control files, and redo logs to a backup location.
 - Restart the database.



- **Hot Backup:** A hot data backup in [Oracle](#) refers to the process of backing up the [database](#) while it is still online and operational also it allows users to access and modify data during the backup process. It is called an online backup. **Hot backups use tools like Oracle Recovery Manager (RMAN) to take backups.** During the hot backup, **Oracle continuously writes changes to the database redo logs.** So those logs are used to fetch the database changes which are made during the backup process.

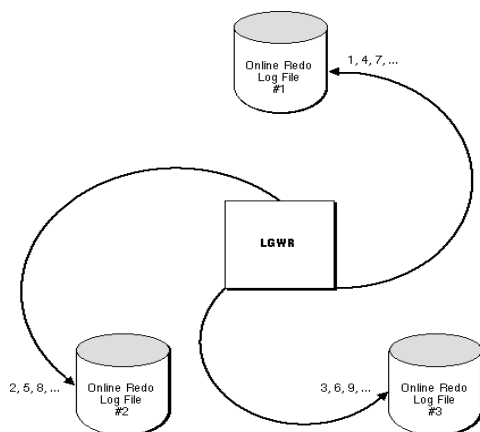
- Put the database in BEGIN BACKUP mode.
- Copy data files while the database is running.
- Use END BACKUP mode after copying.



c. Logical Backups (Data Pump)

- Export a schema: In Oracle, a "logical backup" using Data Pump refers to a backup method that extracts the logical structure and data of database objects like tables, views, procedures, and indexes from a database, storing them in a file format that can be later imported to restore the data, essentially creating a copy of the database schema without capturing the physical file system details; this is achieved using the "Data Pump Export" utility, which is considered a key tool for performing logical backups in Oracle.

Oracle recovery structure and processes



Oracle's recovery structure and processes include backing up and restoring data, rolling forward and rolling back transactions, and recovering from instance failures.

Backup and restore

- Backup: Copies of data files, control files, and redo log files are made
- Restore: Physical files are copied from a backup to their original or new locations
- Recovery: Restored files are updated with changes made to the database after the backup

Rolling forward and rolling back

- Rolling forward: Changes recorded in the redo log are reapplied to the datafiles
- Rolling back: Changes made in uncommitted transactions are rolled back to their original state

Instance recovery

- Oracle automatically recovers a database to its transaction-consistent state after an instance failure
- Instance recovery is performed by a live instance in an Oracle Real Application Clusters configuration

what is physical backup and Logical backup. Also state the difference between them.

A database backup is a copy of storage that is stored on a server. Backup is used to prevent unexpected data loss. If original data gets lost, then with the help of a backup, it is easy to gain access to the data again.

There are two types of database backup.

- Physical backup
- Logical backup

Physical Backup:

Physical database backups are backups of physical files that are used to store and recover databases. These include different data files, control files, archived redo logs, and many more. Typically, physical backup data is kept in the cloud, offline storage, magnetic tape, or on a disc.

There are two methods to perform a physical backup :

1. Operating system utilities
2. Recovery manager

This type of backup is useful when the user needs to restore the complete database in a short period. It is beneficial to provide details of transactions and changes made in databases. It is considered the foundation of the recovery mechanism. This form of backup has the drawback of slowing down database operations.

Advantages:

- It is useful when the user needs to restore the complete database in a short period.
- They provide details of transactions and changes made in databases.

Disadvantage:

- This slows down database operations.

Logical Backup:

It contains logical data which is retrieved from the database. It contains a view, procedure, function, and table. This is useful When users want to restore or transfer a copy of the database to a different location. Logical backups are not as secure as physical backups in preventing data loss. It only provides structural details. Every week, complete logical backups should be performed. Logical backups are used as a supplement to a physical backup.

Advantages:

- This is useful when the user needs to restore the complete database to the last time.
- It was more complex and provides granular recovery capabilities.

Disadvantages:

- Critical for recovery of special components.
- less secure compared to physical backup.
- It only provides structural details.

Physical Backup Vs Logical Backup:

Physical Backup	Logical Backup
Physical database backups are backups of physical files that are used to store and recover databases.	Logical database backups are backups of logical files that are retrieved from the database.
It contains data files, control files, and archived redo logs.	It contains a view, a procedure, a function, and a table.
It copies data files when data is running or stopped.	Using the EXPORT keyword Logical backup is done
A user needs to restore the complete database in a short period of time.	This is useful when users want to restore or transfer a copy of the database to a different location.
More secure than logical backup.	Less secure as compared to Physical backup.

Ques. What are the different types of Failure in Oracle?

Errors and Failures

Several problems can halt the normal operation of an Oracle database or affect database I/O to disk. The following sections describe the most common types. For some of these problems, recovery is automatic and requires little or no action on the part of the database user or database administrator.

User Error

A database administrator can do little to prevent user errors (for example, accidentally dropping a table). Usually, user error can be reduced by increased training on database and application principles. Furthermore, by planning an effective recovery scheme ahead of time, the administrator can ease the work necessary to recover from many types of user errors.

Statement Failure

Statement failure occurs when there is a logical failure in the handling of a statement in an Oracle program. For example, assume all extents of a table (in other words, the number of extents specified in the MAXEXTENTS parameter of the CREATE TABLE statement) are allocated, and are completely filled with data; the table is absolutely full. A valid INSERT

statement cannot insert a row because there is no space available. Therefore, if issued, the statement fails.

If a statement failure occurs, the Oracle software or operating system returns an error code or message. A statement failure usually requires no action or recovery steps; Oracle automatically corrects for statement failure by rolling back the effects (if any) of the statement and returning control to the application. The user can simply re-execute the statement after correcting the problem conveyed by the error message.

Process Failure

A process failure is a failure in a user, server, or background process of a database instance (for example, an abnormal disconnect or process termination). When a process failure occurs, the failed subordinate process cannot continue work, although the other processes of the database instance can.

The Oracle background process PMON detects aborted Oracle processes. If the aborted process is a user or server process, PMON resolves the failure by rolling back the current transaction of the aborted process and releasing any resources that this process was using. Recovery of the failed user or server process is automatic. If the aborted process is a background process, the instance cannot continue to function correctly (usually). Therefore, you must shut down and restart the instance.

Network Failure

When your system uses networks (for example, local area networks, phone lines, and so on) to connect client workstations to database servers, or to connect several database servers to form a distributed database system, network failures (such as aborted phone connections or network communication software failures) can interrupt the normal operation of a database system. For example:

- A network failure might interrupt normal execution of a client application and cause a process failure to occur. In this case, the Oracle background process PMON detects and resolves the aborted server process for the disconnected user process, as described in the previous section.
- A network failure might interrupt the two-phase commit of a distributed transaction. Once the network problem is corrected, the Oracle background process RECO of each involved database server automatically resolves any distributed transactions not yet resolved at all nodes of the distributed database system.

Q. State the Complete and Incomplete Recovery of an Oracle database

Recovery of an Oracle Database

Oracle Database recovery is the process of restoring and applying transactions to bring the database back to a consistent state after a failure. This process is essential for maintaining data integrity and minimizing downtime. Recovery can be categorized into two types: Complete Recovery and Incomplete Recovery.

1. Complete Recovery

Complete recovery is used when all required redo logs (online and archived) are available. This type of recovery ensures that no committed transactions are lost, and the database is restored to the most recent committed state.

When to Use Complete Recovery?

Complete recovery is typically used in the following scenarios:

- **Media Failures:** When a database file (such as a datafile, control file, or redo log file) is lost due to disk corruption or crash.
- **Instance Failures:** When the database crashes due to a power failure, hardware malfunction, or operating system crash.
- **Accidental Deletion of Datafiles:** When a user accidentally deletes a critical database file but has a backup available.
- **Block Corruption:** If a small portion of data is corrupted, and backups along with redo logs are available.

Steps for Performing Complete Recovery

Steps for Performing Complete Recovery

1. **Shut Down the Database (if needed)**
 - If the database is still running but some files are lost or corrupted, shut it down before starting recovery.

SHUTDOWN IMMEDIATE;

2. **Start the Database in Mount Mode**

- The database must be started in MOUNT mode to perform recovery.

STARTUP MOUNT;

3. **Restore the Latest Backup of Datafiles**

- If the datafiles are lost, restore them from a backup.

cp /backup_location/datafile1.dbf /oracle/data/

4. Recover the Database Using Redo Logs

- Apply all redo logs (both online and archived) to restore all transactions.

RECOVER DATABASE;

5. Open the Database in Normal Mode

ALTER DATABASE OPEN;

Advantages of Complete Recovery

- No committed transactions are lost.
- Database returns to the most recent state before the failure.
- Ensures minimal downtime.

Disadvantages of Complete Recovery

- Requires all redo logs to be available.
- If any required redo logs are missing or corrupted, complete recovery is not possible.

2. Incomplete Recovery

Incomplete recovery is used when all redo logs are not available or when the administrator decides to restore the database to an earlier point in time to avoid corruption or accidental data loss. Some transactions will be lost in this type of recovery.

When to Use Incomplete Recovery?

Incomplete recovery is required in the following cases:

- Logical Corruption: If data is logically corrupted due to a software bug or malicious activity.
- Accidental Deletion or Wrong Updates: If a user mistakenly deletes important data or updates the database with incorrect values.
- Missing or Corrupted Redo Logs: If required redo logs for complete recovery are missing, incomplete recovery is the only option.
- Failure of Complete Recovery: If complete recovery is attempted but fails due to missing logs, incomplete recovery is used as a fallback.

Types of Incomplete Recovery

Incomplete recovery can be performed using different techniques depending on how far back in time the database needs to be restored.

1. Time-Based Recovery

Restores the database to a specific point in time before the failure occurred.

2. SCN-Based Recovery

Restores the database to a specific System Change Number (SCN).

3. Cancel-Based Recovery

Allows the user to manually stop the recovery process when enough logs have been applied.

4. Backup Control File Recovery

Used when the control file is lost and must be restored from a backup.

Steps for Performing Incomplete Recovery

1. Shutdown the Database

SHUTDOWN IMMEDIATE;

2. Start the Database in Mount Mode

STARTUP MOUNT;

3. Restore the Necessary Backup Files

```
cp /backup_location/datafile1.dbf /oracle/data/
```

4. Perform the Incomplete Recovery Using One of the Above Methods

```
RECOVER DATABASE UNTIL TIME '2025-02-25:12:00:00';
```

5. Open the Database with RESETLOGS

```
ALTER DATABASE OPEN RESETLOGS;
```

(RESETLOGS is a command used in Oracle Database after performing an incomplete recovery or recovery using a backup control file. It is required because Oracle needs to reset the redo logs and create a new incarnation of the database to prevent inconsistencies.)

Advantages of Incomplete Recovery

- Allows rollback to a known good state, avoiding further corruption.
- Can be performed even if some redo logs are missing.
- Useful in cases of logical corruption or accidental data loss.

Disadvantages of Incomplete Recovery

- Some transactions will be lost.
- Requires a database backup to restore from.
- RESETLOGS must be used, which invalidates previous backups.

Key Differences Between Complete and Incomplete Recovery

Feature	Complete Recovery	Incomplete Recovery
Transaction Loss	No transaction loss	Some transactions lost
Redo Logs Required	All redo logs must be available	Some redo logs can be missing
Use Case	System crash, media failure	Accidental deletion, logical corruption
Database Open Mode	Normal mode	Requires RESETLOGS
SCN or Time-Based Recovery	Not applicable	Possible

Feature	Complete Recovery	Incomplete Recovery
Complexity	Simple	More complex

Export/Import Utilities in Oracle.

Oracle's export and import utilities are essential tools for database administrators, providing a means to back up and restore data efficiently. These utilities are particularly useful for transferring data between databases or within the same database, offering flexibility from the table level to the entire database level.

Export Utility

The export utility in Oracle is designed to extract data from the database, creating a logical backup that can be used to import data into another database or table. This utility is highly flexible, allowing users to specify the level of data extraction, whether it be a full database, a specific schema, or individual tables². For example, in Oracle 10g, the export process is enhanced by the Oracle Data Pump, which provides a faster and more efficient way to handle large volumes of data¹.

Import Utility

The import utility complements the export utility by allowing data to be restored from the backup created by the export process. This utility is crucial for data recovery and migration tasks, enabling the reconstitution of database objects and data in the event of a failure or when moving data to a new environment^{1 2}. The import process can be tailored to restore specific parts of the database, such as particular tables or schemas, making it a versatile tool for database management.

Example

Consider a scenario where a database administrator needs to migrate a database from one server to another. The administrator would first use the export utility to create a backup of the database, specifying the necessary parameters to capture the desired data. Once the export is complete, the backup file can be transferred to the new server. The administrator would then use the import utility to restore the data into the new database environment, ensuring that all necessary data and structures are accurately replicated^{1 5}.

In summary, Oracle's export and import utilities are vital for database management, providing robust solutions for data backup, recovery, and migration. These tools enhance the flexibility and efficiency of database operations, particularly in environments requiring frequent data movement or backup.

Steps to Export Data in Oracle

1. **Choose the Export Utility:** Use Oracle Data Pump for efficient data export. It is a newer and more advanced tool compared to the traditional export utility^{1 2}.
2. **Prepare the Environment:** Ensure that the Oracle environment is correctly set up and that you have the necessary permissions to perform export operations¹.
3. **Execute the Export Command:** Use the **expdp** command to export data. Specify parameters such as the directory, dump file name, and the objects to be exported. For example:

```
expdp username/password DIRECTORY=dir_name DUMPFILE=export.dmp SCHEMAS=schema_name
```

This command exports the specified schema to a dump file.

Steps to Import Data in Oracle

1. **Prepare the Import Environment:** Ensure that the target database is ready to receive the data and that you have the necessary permissions¹.
2. **Execute the Import Command:** Use the **impdp** command to import data. Specify the dump file and other parameters. For example:

```
impdp username/password DIRECTORY=dir_name DUMPFILE=export.dmp SCHEMAS=schema_name
```

This command imports the specified schema from the dump file into the target database.

3. **Verify the Import:** After the import process, verify that the data has been correctly imported by checking the database objects and data integrity.

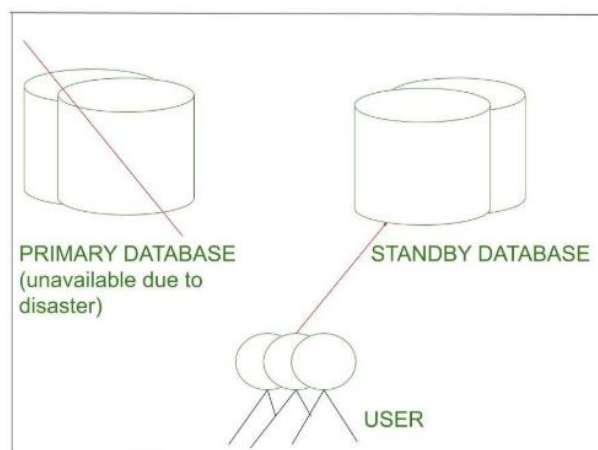
Standby Database

Consider a scenario where primary [database](#) generated is corrupted and what is going to happen in this case? Since database is corrupted, how do we get values and update database by inserting new data? In order to face this situation and maintain the smooth functioning of system, there is concept of **standby database**.

A standby database is copy of database that's been generated from primary database backup. If the primary database is lost or its data becomes corrupted, fail over to standby database can be carried out. Now in such situation, standby database becomes new primary database.

Next question is how data is copied from primary database to standby database. What is the procedure to update standby database in accordance with changes made in primary database?

1. After an interval primary database gets updated. Interval may vary, but primary database contains many updates, such as adding new user, deleting old content, Password change, and so forth. So database updates are captured in redo logs to replicate them to standby database.
2. Updates in redo logs made once are not final. They 're cyclically overwritten. Because redo logs are temporary one, copy of redo log is made. It's called an archive log. Unlike redo log, this is permanent copy.
3. Saved archive log is now moved to standby server. The modifications in transferred archive logs are applied to standby database, thus updating standby database with primary database.



Advantages of StandBy database:

- **Protection of disasters –**
For example, maintaining primary and standby databases so that if the primary database drive fails, the standby database can be activated and normal operations can be resumed and prevent any upcoming disaster.
- **Protection against corruption in data –**
For example, Standby database provides possible protection against incorrect batch jobs, user errors, or application corruption in primary database by not applying corrupted logs to standby database. The uncorrupted standby database can then be activated, making it the primary database.